Grade 11 – Term 1

We practise criterion C9 with worked solutions.

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Open the normal and Student's t reference tables

The standard normal reference is on the first page, and the Student's t reference is on the second page.

This activity evaluates criterion C9.

C9. Students calculate and compare critical values and probability areas from the standard normal and Student's t-distributions, selecting and justifying the appropriate distribution according to whether the population standard deviation is known or estimated from the sample.

1 We prepare to choose and compare distributions.

Find the activities in this section.

1.1. We choose the appropriate distribution.

1.2. We connect critical values with probability areas.

1.3. We use the reference tables for this activity.

1.1 We choose the appropriate distribution.

Assume independent random observations from a normal population with mean μ and standard deviation $\sigma > 0$. These are fixed population parameters; n is sample size, $\bar{X}$ and S are random sample statistics (mean and standard deviation),

and s is the observed value of S.

Population variability	Standardized sample mean	Distribution
σ known	$Z = \dfrac{X - \mu}{\sigma/\sqrt{n}}$	$Z \sim \mathcal{N}(0,1)$
σ unknown, estimated by S	$T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}}$	$T \sim t_\nu, \quad \nu = n-1$

The denominators are the standard error of $\bar{X}$ and its estimate. Both statistics have no units. $\mathcal{N}(0,1)$ is standard normal (mean 0, standard deviation 1); t_ν is Student's t-distribution with ν degrees of freedom (df), requiring $n \geq 2$. Both are continuous and symmetric about zero, with total probability 1; t need not have standard deviation 1.

Estimating σ adds uncertainty: t has heavier tails, hence larger positive cutoffs for the same upper-tail probability below 1/2. As ν increases, t approaches the standard normal distribution. A large sample, or equal numerical values of s and a known σ, does not change their roles or the model choice.



1.2 We connect critical values with probability areas.

For $Z \sim \mathcal{N}(0,1)$ and $T \sim t_\nu$, define the cumulative probability $\Phi(z)$ at a real normal cutoff z, and positive upper-tail critical values by

$$\Phi(z) = P(Z \leq z), \qquad P(Z > z^*_\alpha) = \alpha, \qquad P(T > t^*_{\alpha,\nu}) = \alpha, \quad 0 < \alpha < \tfrac{1}{2}.$$

Write $P(\cdot)$ for probability and $\sim$ for follows the distribution. A cutoff c is a fixed boundary (possibly negative); $\pm c$ denotes both $-c$ and c, and $|Y|$ is the absolute value of Y. For continuous distributions, endpoints do not affect probabilities: $<$ and $\leq$ (or $>$ and $\geq$) are interchangeable. $\approx$ means approximate equality.

For $Y = Z$ or $Y = T$ symmetric about zero, $c > 0$, and $\alpha = P(Y > c)$:

$$P(Y < -c) = \alpha, \qquad P(Y < c) = 1 - \alpha, \quad P(Y > -c) = 1 - \alpha,$$
$$P(|Y| > c) = 2\alpha, \quad P(-c < Y < c) = 1 - 2\alpha.$$

A combined two-tail area q uses $\alpha = q/2$; a central area p uses $\alpha = (1-p)/2$, where p, q are probabilities. The symmetric cutoffs are $\pm c$. For any real endpoints $a < b$,

$$P(a < Y < b) = 1 - P(Y \leq a) - P(Y \geq b), \qquad P(Y > c) = 1 - P(Y \leq c).$$

For normal probabilities, symmetry and cumulative differences give

$$\Phi(-z) = 1 - \Phi(z), \qquad P(Z > z) = 1 - \Phi(z), \qquad P(a < Z < b) = \Phi(b) - \Phi(a).$$



1.3 We use the reference tables for this activity.

Reference page 1: normal cumulative areas. For $z \geq 0$, combine the row (ones and tenths) and column (hundredths) to read $\Phi(z)$, rounded to four decimals; e.g. row 1.2 and column 0.03 give $\Phi(1.23) \approx 0.8907$. For a critical value, seek cumulative area $1 - \alpha$; the hundredth grid does not generally determine a three-decimal cutoff.

Reference page 2: t critical values. Row ν, column α, gives $t^*_{\alpha,\nu}$ to three decimals. Columns are $0.30, 0.20, 0.10, 0.05, 0.025, 0.0$
For adjacent upper-tail columns with $\alpha_1 > \alpha_2$,

$$t^*_{\alpha_1,\nu} < c < t^*_{\alpha_2,\nu} \quad \Longrightarrow \quad \alpha_2 < P(T > c) < \alpha_1.$$

These are probability bounds, not an interpolated estimate. Symmetry, doubling, and complements give other bounds; for example, $1 - 2\alpha_1 < P(-c < T < c) < 1 - 2\alpha_2$. Allow for entry rounding if a cutoff is extremely close to a printed value; the between-entry cutoffs here are well separated from that uncertainty.

Precision and supplements. Report critical values to three decimals and normal areas to four decimals; probabilities at rounded cutoffs are approximate. Use the supplements below for missing precision or rows, without rounding a supplied cutoff or substituting nearby df. The normal critical values are also directly available in the t-sheet's limiting row $\nu = \infty$ (infinite df, the normal limit). Numerical inverse cumulative evaluation gives the same rounded values and supplies the missing t rows. Numerical cumulative evaluation gives $\Phi(1.833) \approx 0.9666$.

Upper-tail probability α	0.10	0.05	0.025	0.01	0.005
Normal z^*_α	1.282	1.645	1.960	2.326	2.576
t: $\nu = 59$	1.296	1.671	2.001	2.391	2.662
t: $\nu = 119$	1.289	1.658	1.980	2.358	2.618

All critical entries above are approximate; a rounded cutoff need not leave exactly its nominal tail probability. Probability bounds use adjacent entries, not numerical interpolation.



2 We compare the standard normal distribution and Student's t-distribution.

Find the activities in this section.

2.1. We compare one-tail, two-tail, and central cutoffs.

2.2. We compare areas at the same cutoff.

2.3. We examine increasing degrees of freedom.

Attempt each task before reading its worked solution.

2.1 We compare one-tail, two-tail, and central cutoffs.

Compare the standard normal distribution with Student's t-distribution having 9 degrees of freedom.

Complete the following tasks.



C9 Find a cutoff leaving an upper-tail area of 0.05, both symmetric cutoffs leaving a combined two-tail area of 0.05, and both cutoffs enclosing the middle 80 percent. Compare the two distributions in each case.

C9 Solution

Use $Z \sim \mathcal{N}(0, 1)$ and $T \sim t_9$.

Request	Upper tail α	z^*_α cutoff(s)	$t^*_{\alpha,9}$ cutoff(s)
(a) One upper tail	0.05	1.645	1.833
(b) Two tails	0.025	$\pm$1.960	$\pm$2.262
(c) Central area	0.10	$\pm$1.282	$\pm$1.383

The cutoffs and differences are approximate. The positive t-cutoff exceeds the corresponding z-cutoff by 0.188, 0.302, and 0.101, respectively. Heavier t-tails require wider symmetric ranges to enclose the same central probability. For the combined tails, divide 0.05 by two; for the central area, divide $1 - 0.80$ by two. Normal critical values use the labeled supplement; Student's values use reference row 9.



2.2 We compare areas at the same cutoff.

Compare the two distributions at the fixed standardized cutoff 1.833.

Complete the following tasks.



C9 Calculate the upper-tail, combined two-tail, and central probabilities for the standard normal distribution and Student's t-distribution with 9 degrees of freedom at symmetric cutoffs of negative 1.833 and positive 1.833. Then give bounds without interpolation for the upper-tail and central probabilities under the same Student's t-distribution at symmetric cutoffs of negative 2.00 and positive 2.00, and compare them with the corresponding normal areas.

C9 Solution

Using the labeled supplemental normal entry, $\Phi(1.833) \approx 0.9666$; for t_9, 1.833 is in column $\alpha = 0.05$. Use $Z \sim \mathcal{N}(0,1)$ and $T \sim t_9$.

Model	Upper tail	Combined tails	Central area
Z	0.0334	0.0668	0.9332
t_9	0.0500	0.1000	0.9000

All areas in the array are approximate: $P(Z > 1.833) \approx 1 - 0.9666 = 0.0334$ and $P(T > 1.833) \approx 0.0500$. Double each upper tail for the combined tails; subtract that result from one for the central area, using symmetry. At 2.00, reference row 9 gives $1.833 < 2.00 < 2.262$, hence

$$0.025 < P(T > 2.00) < 0.05, \qquad 0.90 < P(-2.00 < T < 2.00) < 0.95.$$

For Z, the upper tail is $P(Z > 2.00) \approx 1 - 0.9772 = 0.0228$ and the central area is $P(-2.00 < Z < 2.00) \approx 1 - 2(0.0228) = 0.9544$. Thus the t_9 upper tail is larger and its central area is smaller even using only these bounds.



2.3 We examine increasing degrees of freedom.

Complete the following tasks.



C9 For Student's t-distributions with 5, 19, 59, and 119 degrees of freedom, find the positive cutoff and its negative counterpart enclosing a central area of 0.95. Calculate the difference between each positive cutoff and the positive standard normal cutoff for the same central area. Explain the trend.

C9 Solution

Compare $Z \sim \mathcal{N}(0,1)$ with $T \sim t_\nu$ for each requested ν. The 59 and 119 rows come from the labeled supplement. Central area 0.95 gives $\alpha = 0.025$ and $z^*_{0.025} \approx 1.960$.

ν	$t^*_{0.025,\nu}$	$t^*_{0.025,\nu} - 1.960$
5	2.571	0.611
19	2.093	0.133
59	2.001	0.041
119	1.980	0.020

The symmetric cutoffs are approximately the positive and negative values in each row; differences also use rounded values. Rows 5 and 19 use reference column $\alpha = 0.025$. The decreasing difference shows convergence toward the normal cutoffs, while the finite-ν values remain larger.



3 We apply the models in context.

For every problem, select the distribution and justify it using whether the population standard deviation is known or estimated. Calculate the degrees of freedom when needed, classify the area, and identify the reference row and column, cumulative entry, or labeled supplement used. All sample-mean models satisfy the assumptions in the introduction.

Find the activities in this section.

3.1. We study household expenditure.

3.2. We study investment returns.

3.3. We study delivery times.

3.4. We study production costs.

3.5. We study business revenues.

3.6. We compare wage studies.

3.1 We study household expenditure.

A survey uses 25 monthly household expenditures. The population standard deviation is known to be 120 monetary units. Consider the standardized sample mean.

Complete the following tasks.



C9 Find the cutoff leaving a lower-tail probability of 0.01 and the symmetric cutoffs enclosing the middle 95 percent.

C9 Solution

The standardized sample mean is $Z \sim \mathcal{N}(0,1)$. Use Z because population σ is known; no degrees-of-freedom row is needed. For the left tail, supplemental normal column $\alpha = 0.01$ gives $c \approx -2.326$. For the central area, $\alpha = 0.025$ gives approximate cutoffs ± 1.960.



3.2 We study investment returns.

An analyst samples 10 investment returns. The population standard deviation is unknown and is estimated by a sample standard deviation of 1.8 percentage points. Consider the standardized sample mean.

Complete the following tasks.



C9 Find a positive cutoff exceeded with probability 0.05. Calculate the probability that the standardized sample mean is below negative 1.833 and the probability that it lies between negative 1.833 and positive 2.262.

C9 Solution

The standardized sample mean is $T \sim t_9$. Use t because s estimates unknown σ; $\nu = 10 - 1 = 9$. Row 9, column 0.05, gives $c \approx 1.833$. Symmetry gives $P(T < -1.833) \approx 0.05$. The asymmetric interval excludes tails 0.05 and 0.025, so its area is approximately $1 - 0.05 - 0.025 = 0.925$.



3.3 We study delivery times.

A logistics company samples 15 deliveries. The population standard deviation is unknown; the observed sample standard deviation is 6 minutes. Consider the standardized sample mean.

Complete the following tasks.



C9 Find symmetric cutoffs leaving a combined outside area of 0.10. Then give bounds without interpolation for the probability that the standardized sample mean is below negative 2.30 and for the probability that it is above negative 2.30.

C9 Solution

The standardized sample mean is $T \sim t_{14}$. Use t_{14} because σ is estimated using s and $\nu = 15 - 1 = 14$. The two-tail request gives $\alpha = 0.05$, hence approximate cutoffs ± 1.761. Since $2.145 < 2.30 < 2.624$, the upper-tail area lies between 0.01 and 0.025. By symmetry and complementation,

$$0.01 < P(T < -2.30) < 0.025, \qquad 0.975 < P(T > -2.30) < 0.99.$$



3.4 We study production costs.

A manufacturer samples 20 batch costs from a population with a known standard deviation of 40 monetary units. Consider the standardized sample mean.

Complete the following tasks.



C9 Calculate the probability that the standardized sample mean is above 1.50, the combined probability that it is below negative 1.50 or above positive 1.50, and the probability that it lies between negative 1.50 and positive 1.50. Find the symmetric cutoffs enclosing the middle 98 percent.

C9 Solution

The standardized sample mean is $Z \sim \mathcal{N}(0, 1)$. Use Z because σ is known. Using reference row 1.5, column 0.00, from $\Phi(1.50) \approx 0.9332$, the areas are approximately 0.0668, 0.1336, and 0.8664, respectively. Using the normal critical-value supplement, central area 0.98 gives $\alpha = 0.01$, hence approximate cutoffs ± 2.326.



3.5 We study business revenues.

A study samples 30 businesses and estimates the unknown population standard deviation using a sample standard deviation of 8 thousand monetary units. Consider the standardized sample mean.

Complete the following tasks.

C9 Give bounds without interpolation for the upper-tail, combined two-tail, and central areas at symmetric standardized cutoffs of negative 1.90 and positive 1.90. Compare these areas with those for a second study whose population standard deviation is known, using the same standardized cutoffs.

C9 Solution

The standardized variables are $T \sim t_{29}$ for the first study and $Z \sim \mathcal{N}(0,1)$ for the second. The first study uses t_{29}: σ is unknown and $\nu = 30 - 1 = 29$. Since $1.699 < 1.90 < 2.045$,

$$0.025 < P(T > 1.90) < 0.05, \quad 0.05 < P(|T| > 1.90) < 0.10, \quad 0.90 < P(-1.90 < T < 1.90) < 0.95.$$

The second study uses Z. Its corresponding areas are approximately 0.0287, 0.0574, and 0.9426, from reference row 1.9, column 0.00, where $\Phi(1.90) \approx 0.9713$. These normal areas lie within the coarse t bounds, so the bounds alone do not establish which distribution has the larger tails or smaller central area. The heavier-tail property tells us the t tail areas are larger and its central area smaller.

3.6 We compare wage studies.

Study A samples 6 wages with a known population standard deviation of 200 monetary units. Study B samples 120 wages with an unknown population standard deviation, estimated by a sample standard deviation of 200 monetary units. Consider each standardized sample mean.

Complete the following tasks.



C9 For each study, find the negative standardized cutoff leaving probability 0.025 below it and the symmetric standardized cutoffs enclosing the middle 90 percent. Explain why sample size and the equal numerical standard deviations do not determine the distribution choice.

C9 Solution

Use $Z \sim \mathcal{N}(0,1)$ for A and $T \sim t_{119}$ for B, with $\nu = 120 - 1 = 119$. The 119-row values are supplemental. A uses Z because σ is known, even with $n = 6$: the left cutoff is approximately -1.960 and the central cutoffs are approximately ± 1.645. B uses t_{119} because its standard deviation is an estimate: the left cutoff is approximately -1.980 and the central cutoffs are approximately ± 1.658. A numerical value for s does not supply a known population σ; the large sample only makes the t cutoffs closer to normal values.



4 We practise selecting distributions.

Choose and justify the distribution before calculating. Every requested cutoff and probability refers to the appropriate standardized sample mean, with no measurement units. All samples satisfy the introduction's assumptions. Report cutoffs to three decimals, normal areas to four decimals, and probability ranges without interpolation. Solutions are provided after the complete exercise set.

Find the activities in this section.

4.1. We solve the mixed exercises.

4.2. We check the worked solutions.

4.1 We solve the mixed exercises.

Complete the following tasks.



C9.1 A retail sales model uses 12 observations and a known population standard deviation of 15 monetary units. Find the standardized cutoff leaving probability 0.05 below it, then calculate the probability that the standardized sample mean lies between negative 1.00 and positive 1.50.

C9.2 A price survey uses 12 shops; the unknown population standard deviation is estimated by a sample standard deviation of 15 monetary units. Find symmetric standardized cutoffs leaving a combined outside area of 0.02 and estimate the probability that the standardized sample mean is below negative 2.201.

C9.3 A payroll study selects 5 employees from each of 5 departments. The population standard deviation is unknown and estimated using the sample. Find the degrees of freedom, the standardized cutoff exceeded in 1 percent of outcomes, and the central probability area between negative 2.064 and positive 2.064.

C9.4 An electricity-expenditure model uses 60 households and a known population standard deviation. Find the symmetric standardized cutoffs enclosing a central probability area of 0.99 and calculate the probability that the standardized sample mean is below negative 2.30.

C9.5 A delivery-cost model uses 20 observations and a sample standard deviation to estimate unknown population variability. Give bounds without interpolation for the probability that the standardized sample mean is above 2.30, the combined probability that it is below negative 2.30 or above positive 2.30, and the probability that it lies between negative 2.30 and positive 2.30.

C9.6 Two studies of mean investment returns each use 10 observations. Study A knows the population standard deviation is 2 percentage points; study B estimates the unknown population standard deviation using a sample standard deviation of 2 percentage points. Find the standardized cutoffs enclosing a central area of 0.95 for each. At the fixed standardized cutoff 2.00, calculate or bound each upper-tail area and identify which is larger.

C9.7 A business-cost study uses 6 observations with an unknown population standard deviation estimated from the sample. Find the negative standardized cutoff leaving probability 0.005 below it and estimate the probability that the standardized sample mean lies between negative 2.015 and positive 2.571.

C9.8 A statistical simulation repeatedly samples 60 independent observations from a normal population with unknown standard deviation, estimated separately in each sample. Find standardized cutoffs enclosing the middle 90 percent. Compare their magnitudes with those for a model having known population standard deviation and calculate the difference. Explain whether increasing the sample size to 120 changes the distribution choice.



4.2 We check the worked solutions.

C9.1 – Retail sales

The standardized sample mean is $Z \sim \mathcal{N}(0,1)$. Known σ gives Z. The left-tail cutoff is $c \approx -1.645$ (supplemental normal column $\alpha = 0.05$). Reference rows 1.0 and 1.5, both in column 0.00, give the needed cumulative entries. By symmetry, $\Phi(-1.00) = 1 - \Phi(1.00) \approx 1 - 0.8413 = 0.1587$, so $P(-1.00 < Z < 1.50) \approx 0.9332 - 0.1587 = 0.7745$.



C9.2 – Prices

The standardized sample mean is $T \sim t_{11}$. Unknown σ estimated by s gives t_{11}, with $\nu = 12 - 1 = 11$. Total outside area 0.02 gives $\alpha = 0.01$, so cutoffs are approximately ± 2.718. Row 11, column 0.025, and symmetry give $P(T < -2.201) \approx 0.025$.



C9.3 – Payroll

The standardized sample mean is $T \sim t_{24}$. $n = 5 \times 5 = 25$ and $\nu = 24$. Use t_{24} because population σ is estimated. Column 0.01 gives the approximate positive cutoff 2.492. The value 2.064 is in column 0.025, giving central area approximately $1 - 2(0.025) = 0.95$.



C9.4 – Electricity expenditure

The standardized sample mean is $Z \sim \mathcal{N}(0,1)$. Use Z because σ is known. Using the normal critical-value supplement, central area 0.99 gives $\alpha = 0.005$ and cutoffs ± 2.576. Reference row 2.3, column 0.00, and symmetry give $P(Z < -2.30) = 1 - \Phi(2.30) \approx 1 - 0.9893 = 0.0107$.



C9.5 – Delivery costs

The standardized sample mean is $T \sim t_{19}$. Use t_{19} because s estimates unknown σ and $\nu = 20 - 1 = 19$. The entries $2.093 < 2.30 < 2.539$ give

$$0.01 < P(T > 2.30) < 0.025, \quad 0.02 < P(|T| > 2.30) < 0.05, \quad 0.95 < P(-2.30 < T < 2.30) < 0.98.$$



C9.6 – Investment comparison

Use $Z \sim \mathcal{N}(0,1)$ for A and $T \sim t_9$ for B. A uses Z because σ is known; B uses t_9 because s estimates σ and $\nu = 9$. Central area 0.95 gives $\alpha = 0.025$, hence approximately ± 1.960 for A and ± 2.262 for B. At 2.00, A's upper-tail area is $P(Z > 2.00) = 1 - \Phi(2.00) \approx 1 - 0.9772 = 0.0228$. For B, $1.833 < 2.00 < 2.262$ gives $0.025 < P(T > 2.00) < 0.05$. B's tail is larger, since even its lower bound exceeds 0.0228.



C9.7 – Business costs

The standardized sample mean is $T \sim t_5$. Use t_5 because σ is estimated from the sample; $\nu = 6 - 1 = 5$. The negative cutoff is approximately -4.032, using column 0.005. The asymmetric interval excludes a left tail of 0.05 and a right tail of 0.025, giving area approximately 0.925.



C9.8 – Statistical simulation

The standardized variable is $T \sim t_{59}$ initially and $T \sim t_{119}$ for the larger sample; the known-variability comparison uses $Z \sim \mathcal{N}(0,1)$. Both finite-degree rows used here are supplemental. Use t_{59} because each sample estimates unknown σ; $\nu = 60 - 1 = 59$. Central area 0.90 gives $\alpha = 0.05$, hence approximate cutoffs ± 1.671. Known σ would give approximate Z cutoffs ± 1.645; the approximate magnitude difference is 0.026. With $n = 120$, σ remains unknown, so use t_{119} and approximate cutoffs ± 1.658. Their approximate magnitude difference from the normal cutoffs is 0.013: closer, but the choice remains t.

